Air Quality Intelligence System

Advanced Data Warehouse — Course Report

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1 Team Information

Field	**Value**
Student	Péter Dobosi
Neptun code	DOP3BP
Course	Advanced Data Warehouse
Semester	2025/26 Spring
Repository	`https://github.com/dobosipeter/advanced-data-warehouse-and-database-management-s`
Live system	`https://mw79on-demo.online`
API docs	`https://api.mw79on-demo.online/docs`

2 System Architecture

The system implements an end-to-end data warehouse pipeline: ingesting real-time air quality data from OpenAQ, staging it in raw JSON form, loading it into a normalized OLTP schema, then transforming it into an analytical star schema in the `dw` schema. A machine learning model provides PM2.5 predictions stored back into the warehouse.

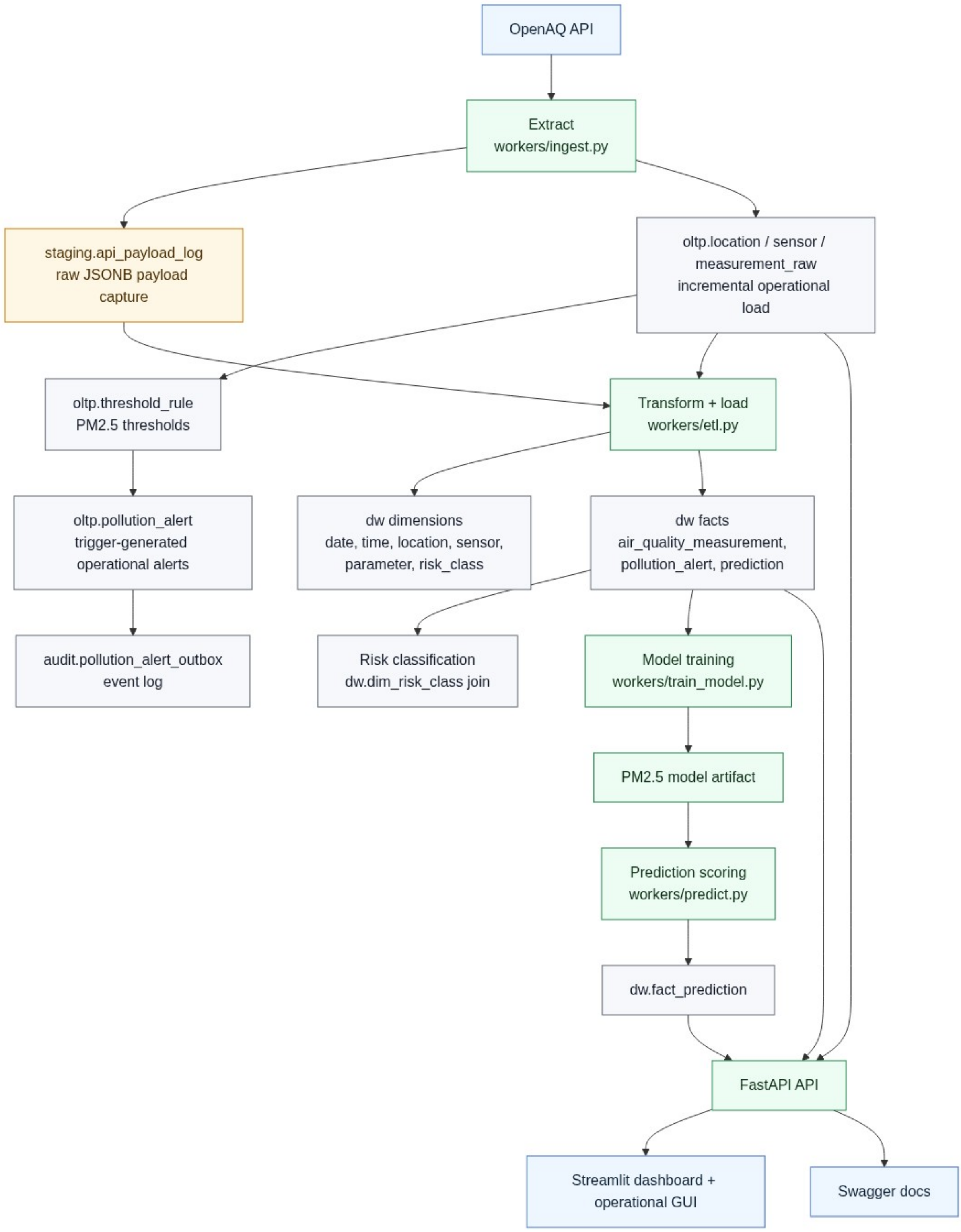

Figure 1: End-to-end data flow: ingestion → staging → OLTP → ETL → Data Warehouse → analytics.

Key data warehouse components:

- **Staging layer** (`staging` schema) — raw JSON API responses preserved for auditability.
- **OLTP layer** (`oltp` schema) — normalized operational tables with triggers and constraints.
- **Data Warehouse** (`dw` schema) — star schema with facts and dimensions.

- **ETL pipeline** — Python script performing incremental extraction, SCD Type 2 dimension updates, and fact loading.
- **ML prediction** — Random Forest model trained on warehouse data, predictions stored as `fact_prediction`.
- **Analytical dashboard** — Streamlit multi-page application with filters, trend charts, heatmaps, and prediction comparisons.

3 Functional Specification

1. **Data Ingestion** — Incremental and bulk ingest from OpenAQ v3, with run logging and error tracking.
2. **ETL Pipeline** — Extracts from OLTP, applies SCD Type 2 logic to location and sensor dimensions, loads facts.
3. **Star Schema** — Three fact tables (measurements, alerts, predictions) with six dimension tables.
4. **SCD Type 2** — Location and sensor dimensions track history with `valid_from/valid_to/is_current` versioning.
5. **Risk Classification** — PM2.5 values classified into Low / Moderate / High / Critical via `dim_risk_class`.
6. **PM2.5 Prediction** — Random Forest model trained on historical DW data; predictions stored in `fact_prediction`.
7. **Automated Refresh** — Orchestration script with modes (full, ingest-etl, predict-only, train-predict) and cron scheduling.
8. **Analytical Dashboard** — Four Streamlit pages: Air Quality Overview, High-Risk Periods, Prediction Insights, Data Operations.
9. **Threshold Alerts** — Configurable alert rules; alerts generated automatically and integrated into the DW as `fact_pollution_alert`.
10. **Deployment** — Docker Compose on Hetzner Cloud VM with Caddy HTTPS, CI/CD via GitHub Actions.

4 Logical Data Model (Star Schema)

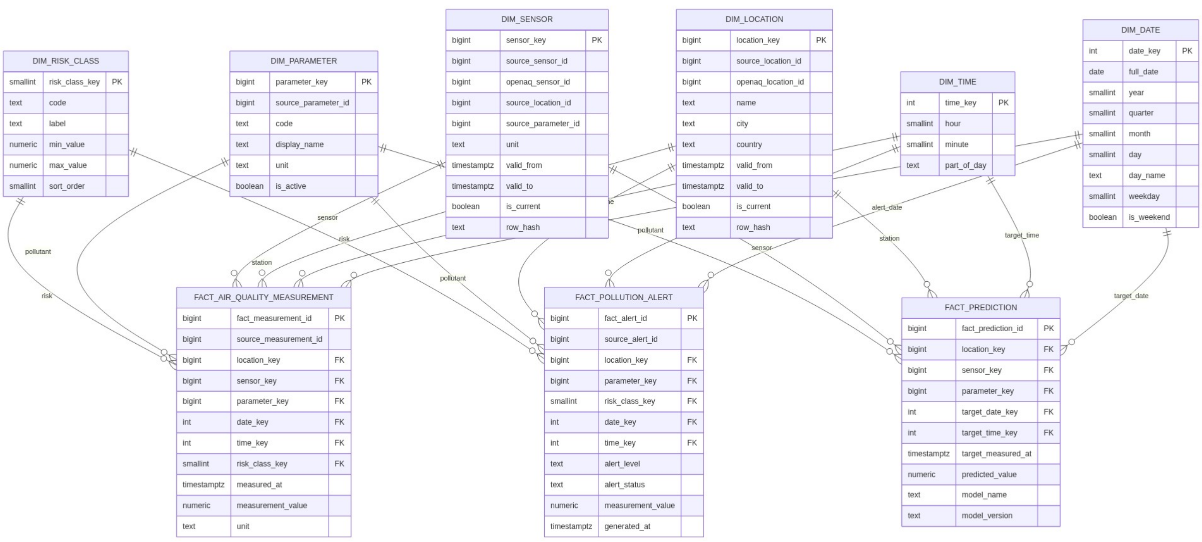

Figure 2: Data warehouse star schema.

Fact Tables

Table	Grain	Measures
`fact_air_quality_measurement`	One reading per sensor per timestamp	`measurement_value`, unit
`fact_pollution_alert`	One alert event	`measurement_value`, level, status
`fact_prediction`	One prediction per location/time	`predicted_value`, model info

Table 1: Fact tables.

Dimension Tables

Table	Type	Description
`dim_date`	Conformed, pre-populated	Calendar attributes (year, quarter, month, weekday, is_weekend)
`dim_time`	Conformed, pre-populated	Hour/minute with part-of-day classification
`dim_parameter`	Type 1	Pollutant definitions (pm25, pm10, no2, o3)
`dim_location`	SCD Type 2	Monitoring stations with validity range
`dim_sensor`	SCD Type 2	Sensors with validity range
`dim_risk_class`	Reference	PM2.5 concentration bands

Table 2: Dimension tables.

5 Technologies Used

Technology	Role
PostgreSQL 16	OLTP + Data Warehouse (single instance, schema separation)
Python 3.12	ETL scripts, ML training, prediction, ingestion
scikit-learn	Random Forest model for PM2.5 prediction
pandas	Data transformation and feature engineering
Streamlit	Analytical multi-page dashboard
Plotly	Interactive charts (line, bar, heatmap)
FastAPI	REST API serving DW data to the frontend
Docker Compose	Service orchestration
Caddy	HTTPS reverse proxy
GitHub Actions	CI/CD deployment pipeline
Hetzner Cloud	Production VM (Ubuntu 24.04)

Table 3: Technology stack.

Notable Data Warehouse Features Demonstrated

- **Star schema** with conformed date/time dimensions
- **SCD Type 2** with hash-based change detection for location and sensor dimensions
- **Incremental ETL** with watermark-based extraction (only new OLTP rows loaded)
- **Risk classification dimension** linking measurements to severity bands
- **ML integration** — predictions stored as facts alongside actuals for comparison
- **Analytical dashboard** with daily aggregation, heatmaps, and error analysis
- **Automated pipeline orchestration** with scheduled cron jobs